

Day 9 - Notes

Component Communication & Data Flow (Salesforce LWC + Aura Concepts)

1. Main Idea of Day 9

Day 9 is about understanding:

- How components talk to each other
 - How data moves in Salesforce applications
 - How UI and backend are connected
 - How modern enterprise systems are structured
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2. Component Communication

Why communication is needed

Applications are built using multiple small components. These components must share data and trigger actions.

Types of communication

1. Parent → Child

- Parent sends data to child
- Used to display or control child component

Example:

- Dashboard → Attendance component
- Account Page → Contact List component

2. Child → Parent

- Child sends event or data to parent
- Parent handles logic or updates UI

Example:

- Form component → Dashboard update
- Attendance component → notify parent

3. Event-Based Communication

- Components communicate using events
- No direct dependency

Advantages:

- Loose coupling
- Reusable components
- Scalable architecture

3. Data Flow in Application (VERY IMPORTANT)

Example: Student Registration Flow

1. UI Layer

- User enters details in form
- Clicks submit

2. Validation Layer

- Frontend checks required fields
- Format validation (email, phone, etc.)

3. Component Layer (LWC/Aura)

- Data is captured in component logic
- Event triggers backend call

4. Apex (Server Side)

- Receives request
- Applies business logic
- Validates data again (security check)

5. Database Layer

- Data stored in Salesforce objects
- Example: Account, Contact, Student

6. Response Layer

- Success or error returned
- UI updates dynamically

4. Dashboard Architecture

Student Dashboard

- Profile component
 - Attendance component
 - Course list component
 - Notification component
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Faculty Dashboard

- Attendance management
 - Marks entry
 - Student tracking
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Admin Dashboard

- User management
 - Course allocation
 - Reports
 - System monitoring
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5. Component Thinking

Each dashboard is made of reusable components:

- Each component has a single responsibility
 - Components can be reused across pages
 - Updates in one component reflect everywhere it is used
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6. Modern vs Legacy Salesforce UI

Visualforce (Old system)

- Page-based structure
 - Heavy and slow
 - Limited flexibility
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Aura Components

- Component-based but complex
- More overhead

- Less efficient
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Lightning Web Components (LWC)

- Modern architecture
 - Faster performance
 - Uses standard web technologies
 - Easy and reusable
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Why Salesforce moved to LWC

- Better performance
 - Cleaner architecture
 - Easier development
 - Industry standard web approach
 - Scalable enterprise design
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7. Frontend vs Backend

Frontend (UI)

Handles:

- User interaction
 - Displaying data
 - Buttons and forms
 - Events
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Backend (Apex)

Handles:

- Business logic

- Database operations
 - Security checks
 - Data processing
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Key Rule

Frontend should not handle heavy logic

Backend should not handle UI

9. Key Learnings

- Components communicate using data and events
- Parent-child communication is core of LWC
- Event-driven architecture improves scalability
- Data flows: UI → validation → Apex → database → response
- Dashboards are built using reusable components
- LWC is modern replacement for Aura and Visualforce
- Modular design is essential in enterprise apps
- Separation of frontend and backend is important